THE BRAUER GROUP OF A C1 FIELD IS ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument. Its inputs are the author's earlier results of the same section, stated in the proof as 'We have seen that'; the counted span contains no label reference and no bibliography citation, so no navigation stub was needed.

Editorial note. The label theorem-C1-brauer-group-zero occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- C1 field (quasi-algebraically closed field). Same chapter, the definition of a C_1 field: every homogeneous form of degree d in more than d variables has a nontrivial zero. Supplies the zero of the reduced norm that the proof produces when the degree d exceeds 1.
- Central simple algebras over a field and their Brauer group. Same chapter, the identification of Br(K) with the set of classes of finite dimensional central division algebras over K. Reduces Br(K) = 0 to the statement that no finite dimensional division algebra with centre K of degree greater than one exists.
- Splitting of a central division algebra by the separable closure and the reduced norm. Same chapter, the fact that D tensor_K K^sep is isomorphic to Mat_d(K^sep) uniquely up to inner isomorphism, so that the determinant descends. Produces the Galois invariant determinant, hence the homogeneous degree d reduced norm map N_red : D -> K used in the counted argument.

Theorem 0.1. *Let K be a C_1 field. Then $Br(K) = 0$.*

Proof. Let D be a finite dimensional division algebra over K with center K. We have seen that

$$D \otimes_K K^{sep} \cong \mathrm{Mat}_d(K^{sep})$$

uniquely up to inner isomorphism. Hence the determinant $\det : \mathrm{Mat}_d(K^{sep}) \to K^{sep}$ is Galois invariant and descends to a homogeneous degree d map

$$\det = N_{\mathrm{red}} : D \longrightarrow K$$

called the *reduced norm*. Since K is C_1, if $d > 1$, then there exists a nonzero $x \in D$ with $N_{\mathrm{red}}(x) = 0$. This clearly implies that x is not invertible, which is a contradiction. Hence $\mathrm{Br}(K) = 0$. □

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